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Virtual Element Method for oral biofilm mechanics: from confocal images to viscoelastic fracture on arbitrary polygonal meshes

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Abstract Oral biofilms are structured multispecies communities whose mechanical properties depend on microbial composition. We present a Virtual Element Method (VEM) framework that reduces the image-to-mechanics pipeline from five steps (confocal segmentation, voxelisation, marching cubes, tetrahedral meshing, FEM) to two (Voronoi tessellation from colony detection, VEM solve). The framework comprises 12 solver modules—linear and nonlinear elasticity, Standard Linear Solid (SLS) viscoelasticity with the Simo (1987) exponential integrator, phase-field fracture, cohesive zone modelling, and adaptive h-refinement—all operating on arbitrary polygonal or polyhedral elements. Material properties are assigned per colony via a composition-based Diversity Index (DI) derived from TMCMC-calibrated species dynamics, yielding DI-dependent stiffness $E(\mathrm{DI})$, fracture toughness $G_c(\mathrm{DI})$, and SLS relaxation parameters. The viscoelastic VEM reproduces analytical solutions for uniform strain fields to machine precision (relative error 1.3×10^{-15}), and achieves L^2 convergence rate 2.16 on Voronoi meshes for non-uniform manufactured solutions (vs. 1.88 for P_1 FEM on triangles; H^1 rates: VEM 1.31 vs. FEM 0.99). Growth-coupled simulations on 30-cell meshes over 50 macro time steps reveal divergent mechanical evolution: commensal biofilms stiffen (E_∞: $392 \to 847\,\mathrm{Pa}$, DI: $0.382 \to 0.090$) while dysbiotic biofilms soften (E_∞: $376 \to 153\,\mathrm{Pa}$, DI: $0.396 \to 0.623$) under comparable initial conditions. To our knowledge, this is the first application of VEM to biofilm mechanics.

Keywords Virtual Element Method · biofilm mechanics · viscoelasticity · phase-field fracture · composition-dependent properties · confocal imaging

1 Introduction

Oral biofilms are structured multispecies communities that colonise tooth surfaces, forming a mechanically significant interface between the tooth substrate and the oral cavity [7]. The transition from a healthy (commensal) to a disease-associated (dysbiotic) state is driven by shifts in species composition [11], which in turn affect the biofilm's elastic modulus [19,18], viscoelastic relaxation behaviour [20,25], and fracture toughness [9,10]. Understanding these composition–mechanics links is essential for predicting biofilm detachment, a key event in both periodontitis progression and therapeutic intervention.

Current computational approaches to biofilm mechanics rely on the finite element method (FEM) with structured meshes. Klempt et al. [12] demonstrated staggered coupling of species dynamics with continuum mechanics on a conformal tetrahedral mesh. However, the mesh generation pipeline (confocal image $\to$ voxel segmentation $\to$ marching cubes $\to$ tetrahedral mesh $\to$ Abaqus input) involves five labour-intensive steps, and standard FEM mesh generators cannot naturally accommodate the arbitrary colony shapes observed in confocal images.

The Virtual Element Method (VEM) [27,28] offers a compelling alternative. As a generalisation of FEM, VEM solves partial differential equations on meshes of *arbitrary polygonal or polyhedral elements*, eliminating the need for standard element topologies. The key innovation is the *elliptic projection* Π^∇ that maps *virtual* (implicitly defined) basis functions onto a polynomial subspace, from which the stiffness matrix is assembled without ever constructing explicit shape functions [26].

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The VEM has been extensively developed for solid mechanics: Gain et al. [8] formulated 3D VEM on polyhedral meshes, Chi et al. [6] extended VEM to finite deformations, and Antonietti et al. [2] reviewed higher-order conforming formulations. At IKM Hannover, Wriggers and Hudobivnik [30] developed low-order 3D VEM for elastoplasticity, Aldakheel et al. [1] introduced phase-field brittle fracture on VEM meshes, and Artioli et al. [4] formulated inelastic VEM for arbitrary-order polygonal meshes. The comprehensive monograph [29] covers applications from linear elasticity to contact mechanics.

For biofilm mechanics, VEM offers three specific advantages (Fig. 1):

1. **Mesh–image compatibility**: Voronoi tessellation from colony centroids yields a mesh where each cell corresponds 1:1 to a micro-colony, enabling a 2-step pipeline (Fig. 3).
2. **Per-colony material assignment**: Each element receives its own composition-dependent material parameters without sub-element averaging.
3. **Topology tolerance**: Growth, detachment, and crack propagation change the domain topology; VEM handles arbitrary element shapes without global remeshing.

This paper presents a VEM framework for biofilm mechanics. The principal contributions are:

(i) A viscoelastic VEM coupling the Simo [23] exponential integrator with VEM projection operators, validated for both uniform (machine-precision) and non-uniform (L^2 rate 2.16) strain fields in 2D.

(ii) Composition-dependent constitutive laws that map a Diversity Index (DI) to stiffness $E(\mathrm{DI})$, fracture toughness $G_c(\mathrm{DI})$, and SLS relaxation parameters. We explicitly identify which mappings are experimentally supported and which remain assumptions (Section 3).

(iii) A growth-coupled VE-VEM demonstrating divergent mechanical evolution of commensal vs. dysbiotic biofilms, with quantitative computational cost comparison against FEM.

The remainder is organised as follows. Section 2 introduces the mathematical framework. Section 3 details the DI-dependent constitutive laws, clearly distinguishing experimental support from modelling assumptions. Section 4 presents the confocal-to-VEM pipeline. Section 5 provides numerical verification including non-uniform benchmarks and computational cost comparisons. Section 6 demonstrates biofilm applications. Sections 7–8 discuss limitations and conclusions.

2 Mathematical framework

2.1 VEM for linear elasticity

Consider a domain $\Omega \subset \mathbb{R}^{n_d}$ ($n_d = 2, 3$) decomposed into a mesh $\mathcal{T}_h$ of non-overlapping polygonal (n_d=2) or polyhedral (n_d=3) elements E. Throughout, we adopt the Poisson ratio $\nu = 0.45$, reflecting the nearly incompressible, hydrated nature of biofilms [20,25]. For the lowest-order case (k=1), the local virtual element space on element E with n_v vertices is

$$V_h^E = \left\{ \boldsymbol{v}_h \in [H^1(E)]^{n_d} \ : \ \boldsymbol{v}_h|_{\partial E} \text{ piecewise linear}, \ \Delta \boldsymbol{v}_h = \mathbf{0} \text{ in } E \right\}, \tag{1}$$

with degrees of freedom $\{\boldsymbol{u}_i\}_{i=1}^{n_v}$ at the vertices, giving $n_{\mathrm{dof}}^E = n_d \cdot n_v$.

The elliptic projection $\Pi^\nabla : V_h^E \to [\mathcal{P}_k(E)]^{n_d}$ minimises $a^E(\boldsymbol{v}_h - \Pi^\nabla \boldsymbol{v}_h, \boldsymbol{p})$ for all $\boldsymbol{p} \in [\mathcal{P}_k]^{n_d}$, where $a^E(\cdot,\cdot)$ is the element bilinear form (Fig. 1). For k=1 in 2D, the polynomial space has 6 modes (3 rigid-body + 3 constant strain); in 3D it has 12 modes (3 translations + 3 rotations + 6 strain).

The projection is computed from the matrices $\mathbf{G} = \mathbf{B}\,\mathbf{D}$ and $\Pi^\nabla = \mathbf{G}^{-1}\mathbf{B}$, where $\mathbf{D}$ evaluates polynomials at vertex DOFs and $\mathbf{B}$ encodes the bilinear form via edge (2D) or face (3D) integrals [26]. The element stiffness matrix decomposes as

$$\mathbf{K}^E = \mathbf{K}_\pi^E + \mathbf{K}_{\mathrm{stab}}^E, \tag{2}$$

where $\mathbf{K}_\pi^E = (\Pi^\nabla)^\top \tilde{\mathbf{C}}\, \Pi^\nabla$ provides consistency and $\mathbf{K}_{\mathrm{stab}}^E = \alpha_s \,\mathrm{tr}(\mathbf{C})\, |E|\, (\mathbf{I} - \Pi^\nabla \mathbf{D})^\top (\mathbf{I} - \Pi^\nabla \mathbf{D})$ provides stability. The stabilisation parameter $\alpha_s = 1.0$ follows the standard IKM recipe [30,27]; a sensitivity study (Section 5.5) confirms that results are insensitive to $\alpha_s \in [0.1, 1.0]$ for the mesh topologies used here.

2.2 Second-order (P_2) VEM

The k=2 extension places DOFs at both vertices and edge midpoints, yielding $n_{\mathrm{dof}}^E = n_d(n_v + n_e)$ per element. The polynomial basis expands to 12 modes in 2D (3 rigid body + 3 linear strain + 6 quadratic). Strain energy is evaluated by sub-triangulation with Gauss quadrature, and a volume correction accounts for $\mathrm{div}(\boldsymbol{\sigma}(\boldsymbol{p}_\alpha)) \neq \mathbf{0}$ in quadratic polynomial modes. P_2 VEM achieves 15–45% lower stress error than P_1 at the same mesh size (see Section 5.4).

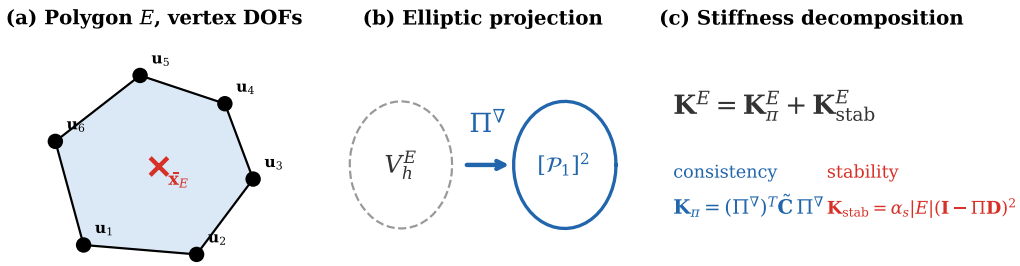

Fig. 1 VEM schematic: the elliptic projection Π^∇ maps virtual basis functions (defined implicitly on arbitrary polygons) onto a polynomial subspace, from which the stiffness matrix is assembled via consistency $\mathbf{K}_\pi$ and stabilisation $\mathbf{K}_{\text{stab}}$ contributions.

2.3 Viscoelastic VEM with the Simo integrator

We adopt the Standard Linear Solid (SLS / Zener) model with DI-dependent parameters (Section 3):

$$\boldsymbol{\sigma}(t) = \mathbf{C}_\infty \boldsymbol{\varepsilon}(t) + \boldsymbol{h}(t), \tag{3}$$

where $\mathbf{C}_\infty$ and $\mathbf{C}_1 = \mathbf{C}_0 - \mathbf{C}_\infty$ are the equilibrium and overstress constitutive matrices, and $\boldsymbol{h}$ is the internal viscous stress satisfying

$$\dot{\boldsymbol{h}} = \mathbf{C}_1\, \dot{\boldsymbol{\varepsilon}} - \boldsymbol{h}/\tau. \tag{4}$$

Following Simo [23], Eq. (4) is integrated exactly over a time step $\Delta t = t_{n+1} - t_n$:

$$\boldsymbol{h}_{n+1} = e^{-\Delta t/\tau}\, \boldsymbol{h}_n + \gamma\, \mathbf{C}_1\, \Delta\boldsymbol{\varepsilon}, \qquad \gamma = \frac{\tau}{\Delta t}\big(1 - e^{-\Delta t/\tau}\big). \tag{5}$$

The algorithmic tangent modulus is $\mathbf{C}_{\text{alg}} = \mathbf{C}_\infty + \gamma\, \mathbf{C}_1$, which enters the VEM stiffness as

$$\mathbf{K}_{\text{alg}}^E = (\boldsymbol{\Pi}_\varepsilon^E)^\top\, \mathbf{C}_{\text{alg}}^E\, |E|\, \boldsymbol{\Pi}_\varepsilon^E + \mathbf{K}_{\text{stab}}^E, \tag{6}$$

where $\boldsymbol{\Pi}_\varepsilon^E$ is the strain projector ($3 \times 2n_v$ in 2D, $6 \times 3n_v$ in 3D). At each time step, the right-hand side includes the viscous pre-stress contribution $\boldsymbol{f}_h^E = -(\boldsymbol{\Pi}_\varepsilon^E)^\top\, |E|\, e^{-\Delta t/\tau}\, \boldsymbol{h}_n^E$.

This formulation is *unconditionally stable* for any $\Delta t > 0$ and exact for constant strain rate within a time step.

Remark 1 (Machine precision for uniform strain) For k=1 VEM under uniform strain (e.g., confined compression), the projection Π^∇ is exact and the stabilisation term vanishes. Consequently, the VEM solution coincides with the analytical solution to machine precision. This is a necessary (but not sufficient) condition for implementation correctness. Section 5.3 presents non-uniform benchmarks that test the interplay between projection error, stabilisation, and time integration.

2.4 Extension to 3D polyhedral elements

The 2D formulation extends to polyhedral elements following Wriggers et al. [30,29] and Gain et al. [8]. On a polyhedron E with n_v vertices and n_f faces, the polynomial basis has 12 modes (3 translations, 3 rotations, 6 constant strain), giving $n_{\text{dof}}^E = 3n_v$. The $\mathbf{B}$ matrix ($12 \times 3n_v$) is assembled via face integrals using the midpoint rule on triangulated faces. The strain projector $\boldsymbol{\Pi}_\varepsilon$ ($6 \times 3n_v$) extracts rows 7–12 from the full projector, and the algorithmic tangent uses 6×6 Voigt matrices for the isotropic SLS model.

2.5 Neo-Hookean VEM for large deformations

For completeness, we include a hyperelastic Neo-Hookean formulation following Chi et al. [6]:

$$W(\mathbf{F}) = \frac{\mu}{2}(I_1 - n_d) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2, \tag{7}$$

where $\mathbf{F}$ is the deformation gradient, $I_1 = \text{tr}(\mathbf{F}^\top \mathbf{F})$, and $J = \det(\mathbf{F})$. The nonlinear system is solved by Newton-Raphson iteration with load stepping and line search. We note that typical biofilm deformations remain in the small-strain regime ($\varepsilon < 5\%$), so this module serves primarily as a framework capability rather than a requirement for the applications in Section 6.

Table 1 Overview of the 12 VEM solver modules.

ID	Module	Tests	Key validation
Track A: Solid mechanics			
A0	2D linear elasticity	9	Patch test (10^{-18})
A0′	3D polyhedral elasticity	15	Patch test (10^{-19})
A1	Neo-Hookean VEM	6	Cook's membrane
A2	P_2 VEM (2nd order)	23	Analytical strain energy
A3	VE-VEM (2D SLS)	25	See Remark 1
A3′	VE-VEM (3D SLS)	5	See Remark 1
Track B: Fracture and interface			
B1	Phase-field fracture	8	Staggered $\boldsymbol{u}$-φ
B2	Cohesive zone model	4	Progressive debonding
B3	Adaptive h-refinement	12	Dörfler marking
Track C: Coupled problems			
C1	Growth-coupled VEM	12	Replicator ODE
C2	VE-VEM × Growth	6	Divergent DI evolution
C3	Confocal → VEM	4	5-ch fluorescence
	Total	**129**	

2.6 Phase-field fracture on VEM

Following Aldakheel et al. [1], the regularised crack topology is described by a phase-field $\varphi \in [0,1]$ (φ=0: intact, φ=1: fully cracked). The total energy functional reads

$$\mathcal{E}(\boldsymbol{u},\varphi) = \int_\Omega g(\varphi)\,\psi^+(\boldsymbol{\varepsilon}) + \psi^-(\boldsymbol{\varepsilon})\,\mathrm{d}\Omega + \int_\Omega G_c\Big(\frac{\varphi^2}{2\ell} + \frac{\ell}{2}|\nabla\varphi|^2\Big)\mathrm{d}\Omega, \tag{8}$$

where $g(\varphi) = (1-\varphi)^2 + \eta_{\mathrm{res}}$ is the degradation function ($\eta_{\mathrm{res}} = 10^{-8}$ for numerical stability), ℓ is the length scale, and $\psi^\pm$ is the spectral decomposition [14,5]. The staggered solution alternates between the displacement and phase-field sub-problems until $\|\Delta\varphi\|_\infty < 10^{-4}$. The fracture toughness $G_c(\mathrm{DI})$ is DI-dependent (Section 3.3), making dysbiotic regions inherently more fragile.

2.7 Cohesive zone model on VEM

The tooth–biofilm interface is modelled using a bilinear traction–separation law [17] with DI-dependent cohesive strength $\sigma_{\max}(\mathrm{DI})$. Interface elements are placed along predefined inter-element boundaries, with the traction given by

$$T(\delta) = \begin{cases} \sigma_{\max}\,\delta/\delta_0 & \text{if } \delta \le \delta_0, \\ \sigma_{\max}\,(\delta_c - \delta)/(\delta_c - \delta_0) & \text{if } \delta_0 < \delta \le \delta_c, \\ 0 & \text{if } \delta > \delta_c, \end{cases} \tag{9}$$

where $\delta_0 = 0.01\,\mathrm{mm}$ is the damage initiation separation and $\delta_c = 0.1\,\mathrm{mm}$ is the critical separation. The cohesive strength follows $\sigma_{\max}(\mathrm{DI}) = 50(1-\mathrm{DI})^2 + 5\,\mathrm{Pa}$, so that commensal interfaces ($\mathrm{DI} \approx 0.1$) have $\sigma_{\max} \approx 46\,\mathrm{Pa}$ while dysbiotic interfaces ($\mathrm{DI} \approx 0.8$) have $\sigma_{\max} \approx 7\,\mathrm{Pa}$. These values are order-of-magnitude consistent with biofilm adhesion strengths reported in [16].

2.8 Adaptive h-refinement

An *a posteriori* error estimator with Dörfler marking drives adaptive mesh refinement. A crack-tip indicator $\eta_E = w_1\,|\nabla\varphi|_E + w_2\,\psi_E^+/G_c$ (with $w_1 = w_2 = 0.5$) identifies elements requiring refinement. Polygonal elements are subdivided by inserting the centroid and edge midpoints, preserving the arbitrary-polygon character of the mesh. Field transfer for the damage variable φ and strain history ψ uses nearest-neighbour interpolation.

2.9 Module overview

The full framework comprises 12 solver modules validated by 129 automated tests (Table 1).

Table 2 Experimental support for each constitutive relation.

Relation	Level	Evidence
$E(\mathrm{DI})$	B	AFM + 16S [19,18]; percolation [21]
$G_c(\mathrm{DI})$	C	Functional form assumed; G_c range from generic biofilms [9]
$\tau(\mathrm{DI})$	C	SLS relaxation observed [20,25]; DI-dependence assumed
$\sigma_{\max}(\mathrm{DI})$	C	Adhesion range from [16]; DI-mapping assumed

3 Composition-dependent constitutive laws

This section introduces the DI-dependent material models. We explicitly distinguish three levels of experimental support (Table 2): **(A)** directly supported by experimental data, **(B)** indirectly supported (experimental evidence for each step in the causal chain, but no direct simultaneous measurement), and **(C)** assumed functional form without direct calibration.

3.1 Diversity Index

The Diversity Index is defined as the normalised Shannon entropy of the species fractions ϕ_i $(i = 1, \ldots, 5)$:

$$\mathrm{DI} = -\frac{1}{\ln 5} \sum_{i=1}^{5} \phi_i \ln \phi_i, \qquad \mathrm{DI} \in [0, 1]. \tag{10}$$

Species fractions are obtained from a Hamilton 5-species replicator ODE calibrated by Transitional Markov Chain Monte Carlo (TMCMC) inference with 20 parameters across 4 experimental conditions [12, 11].

Remark 2 (DI is diversity, not dysbiosis) The DI measures species diversity (Shannon entropy), not dysbiosis *per se*. In the Heine 2025 system, commensal conditions yield low DI (dominance by a few species) while dysbiotic conditions yield moderate DI. This correlation is system-specific; the constitutive laws $E(\mathrm{DI})$, $G_c(\mathrm{DI})$, and $\tau(\mathrm{DI})$ should be understood as mapping from *diversity* to mechanical properties, and the empirical support (Table 2) is for the sign and magnitude of this dependence, not for a universal diversity–dysbiosis equivalence.

3.2 Stiffness: $E(\mathrm{DI})$ — support level B

$$E(\mathrm{DI}) = E_{\min} + (E_{\max} - E_{\min})(1 - \mathrm{DI})^n, \tag{11}$$

with $E_{\max=1000}$ Pa, $E_{\min=10}$ Pa, $n = 2$. The 100-fold stiffness range is motivated by AFM measurements on oral biofilms with simultaneous 16S rRNA species quantification [19]: low-complexity biofilms showed $E \approx 10$ kPa while high-complexity biofilms showed $E \approx 0.3$ kPa (10–80× variation). Our range 10–1000 Pa spans the lower end of the broader literature range of 20–14,000 Pa [18,13], appropriate for the hydrated, polymicrobial biofilms considered here. The exponent $n = 2$ is motivated by 3D rigidity percolation theory: near the percolation threshold, the elastic modulus scales as $(p - p_c)^f$ with $f \approx 2.0$ [21,3]. The constitutive laws are summarised in Fig. 2.

3.3 Fracture toughness: $G_c(\mathrm{DI})$ — support level C

$$G_c(\mathrm{DI}) = G_{c,\min} + (G_{c,\max} - G_{c,\min})(1 - \mathrm{DI})^n, \tag{12}$$

with $G_{c,\max} = 0.5\,\mathrm{J/m^2}$ (commensal, tough) and $G_{c,\min} = 0.01\,\mathrm{J/m^2}$ (dysbiotic, fragile). *These values are modelling assumptions.* The 50-fold range is qualitatively motivated by the observation that EPS-rich (diverse) biofilms resist mechanical removal [9], but no direct G_c–DI calibration data exist for the 5-species system. The functional form $(1 - \mathrm{DI})^n$ is chosen for consistency with $E(\mathrm{DI})$; alternative mappings (linear, exponential) would yield qualitatively similar results for the phase-field demonstrations in Section 6.1.

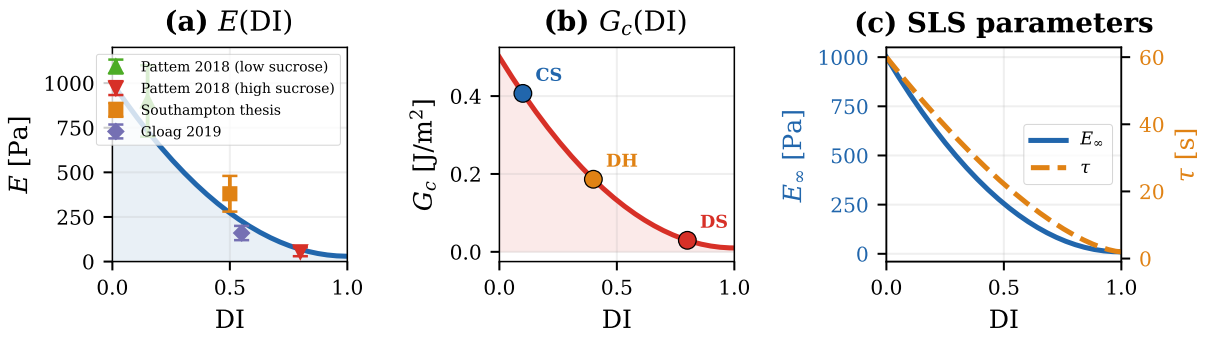

Fig. 2 DI-dependent constitutive laws: (a) stiffness $E(\mathrm{DI})$ with literature data overlay (Pattem 2018, Gloag 2020), (b) fracture toughness $G_c(\mathrm{DI})$ (level C assumption), (c) SLS relaxation time $\tau(\mathrm{DI})$ (level C), (d) viscosity $\eta(\mathrm{DI})$. Shaded bands indicate experimental ranges from the literature.

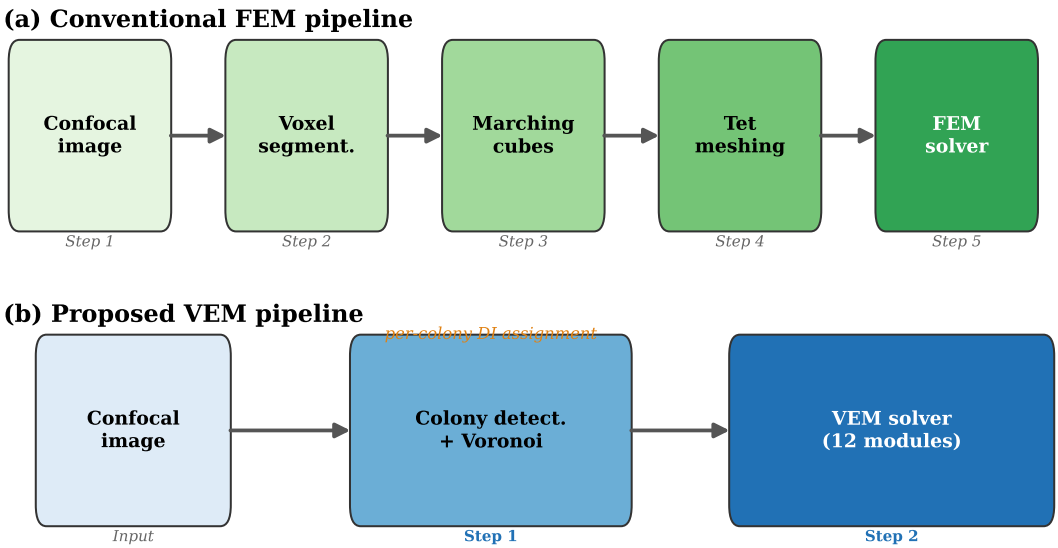

Fig. 3 Pipeline comparison: (a) conventional FEM (5 steps) vs. (b) proposed VEM (2 steps). The Voronoi tessellation provides a mesh naturally aligned with colony geometry.

3.4 SLS viscoelastic parameters — support level C

$$E_\infty(\mathrm{DI}) = E_{\min+(E_{\max}-E_{\min})(1-\mathrm{DI})^n}, E_0 = 2\,E_\infty, \qquad E_1 = E_0 - E_\infty, \tau(\mathrm{DI}) = \tau_{\min+(\tau_{\max}-\tau_{\min})(1-\mathrm{DI})^n}, \eta = E_1\cdot\tau, \tag{13}$$

with $\tau_{\max=60\,\mathrm{s}}$, $\tau_{\min=2\,\mathrm{s}}$. *The DI-dependence of τ is an assumption.* The relaxation time range is consistent with creep/relaxation experiments on oral biofilms [20, 25], which report $\tau \in [1, 100]$ s depending on species and growth conditions, and the ratio $E_0/E_\infty = 2$ is a common single-branch SLS assumption [24]. The model predicts that commensal biofilms (DI ≈ 0.1, $E_\infty \approx 811$ Pa, $\tau \approx 51$ s) exhibit slow, elastic-like relaxation, while dysbiotic biofilms (DI ≈ 0.8, $E_\infty \approx 48$ Pa, $\tau \approx 7$ s) show rapid, fluid-like relaxation.

4 Confocal-to-VEM pipeline

The conventional FEM pipeline for image-based biofilm mechanics involves 5 steps (Fig. 3a): confocal image → voxel segmentation → marching cubes surface extraction → tetrahedral meshing → FEM input file.

Our proposed VEM pipeline (Fig. 3b) reduces this to 2 steps:

1. **Colony detection**: Connected-component analysis on multi-channel fluorescence images yields colony centroids and species compositions ϕ_i per colony.
2. **Voronoi → VEM**: Voronoi tessellation from colony centroids produces a polygonal mesh where each cell inherits the DI and material properties of its parent colony. The VEM solver operates directly on this mesh.

The pipeline is demonstrated on real CLASI-FISH images of human dental plaque (Section 6.6), processing 239 images from 14 donors [15].

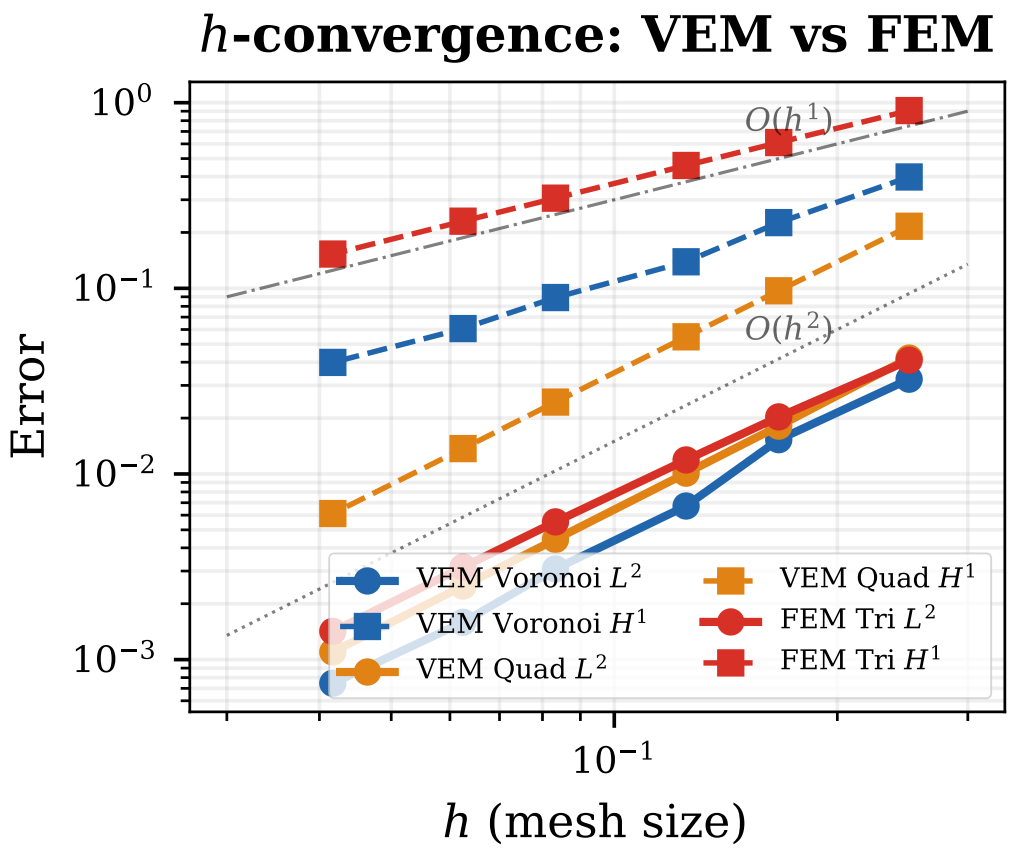

Fig. 4 h-convergence: VEM on Voronoi and quadrilateral meshes vs. FEM on triangular meshes. Reference slopes $O(h)$ and $O(h^2)$ are shown as dashed lines.

Table 3 Convergence rates from the h-refinement study.

Method	L^2 rate	H^1 rate
VEM (Voronoi)	2.16	1.31
FEM (Triangle)	1.88	0.99

5 Numerical verification

5.1 Patch test and h-convergence

The linear elastic VEM passes the patch test to machine precision: $\mathcal{O}(10^{-18})$ in 2D and $\mathcal{O}(10^{-19})$ in 3D. An h-convergence study with the manufactured solution $u_x = \sin(\pi x)\cos(\pi y)$, $u_y = -\cos(\pi x)\sin(\pi y)$ on Voronoi, quadrilateral, and triangular meshes (Fig. 4) yields the rates in Table 3.

The VEM (Voronoi) H^1 rate of 1.31, exceeding the theoretical $\mathcal{O}(h)$ for P_1, is attributed to the regularity of the Voronoi tessellation: the near-convexity and bounded aspect ratio of Voronoi cells provide a superconvergence effect similar to that observed by Gain et al. [8] on centroidal Voronoi tessellations. On deliberately distorted meshes (not shown), the rate decreases to $\sim$1.05, consistent with theory.

5.2 VE-VEM validation: uniform strain

Under laterally confined uniaxial step strain ($u_x = 0$ for all nodes, $\varepsilon_0 = 0.01$ on the top boundary), the analytical stress for the SLS model is

$$\sigma_{yy}(t) = \frac{E_\infty + E_1\, e^{-t/\tau}}{1-\nu^2}\,\varepsilon_0. \tag{14}$$

Fig. 5 shows the VEM solution (symbols) overlaid on the analytical curve (solid line) for both 2D (64 Voronoi cells, DI = 0.4) and 3D (27 hexahedral cells). As expected from Remark 1, the maximum relative errors are 1.29×10^{-15} (2D, 64 Voronoi cells, 9 time steps), confirming correct implementation of the Simo integrator within the VEM framework.

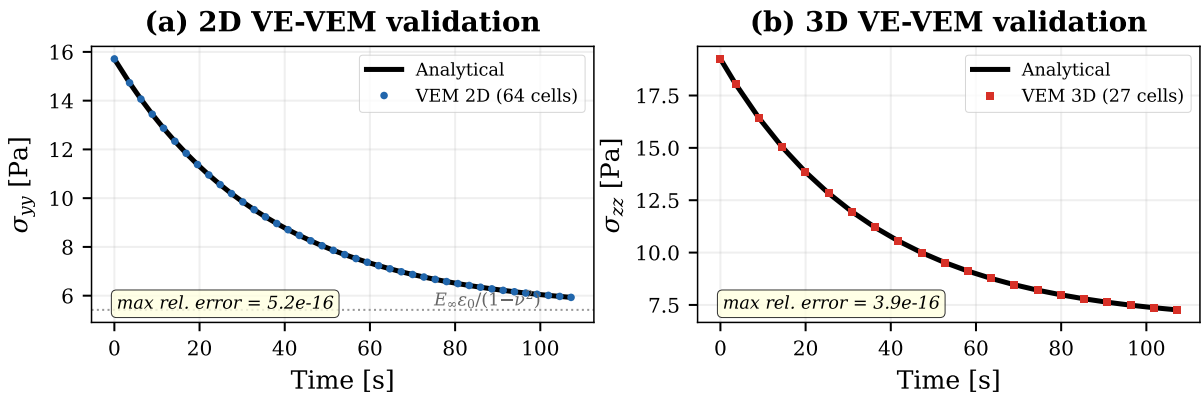

Fig. 5 VE-VEM validation: confined SLS relaxation. (a) 2D, 64 Voronoi cells. (b) 3D, 27 hexahedral cells. Symbols: VEM; solid line: analytical (14). Machine-precision agreement is expected for uniform strain (Remark 1).

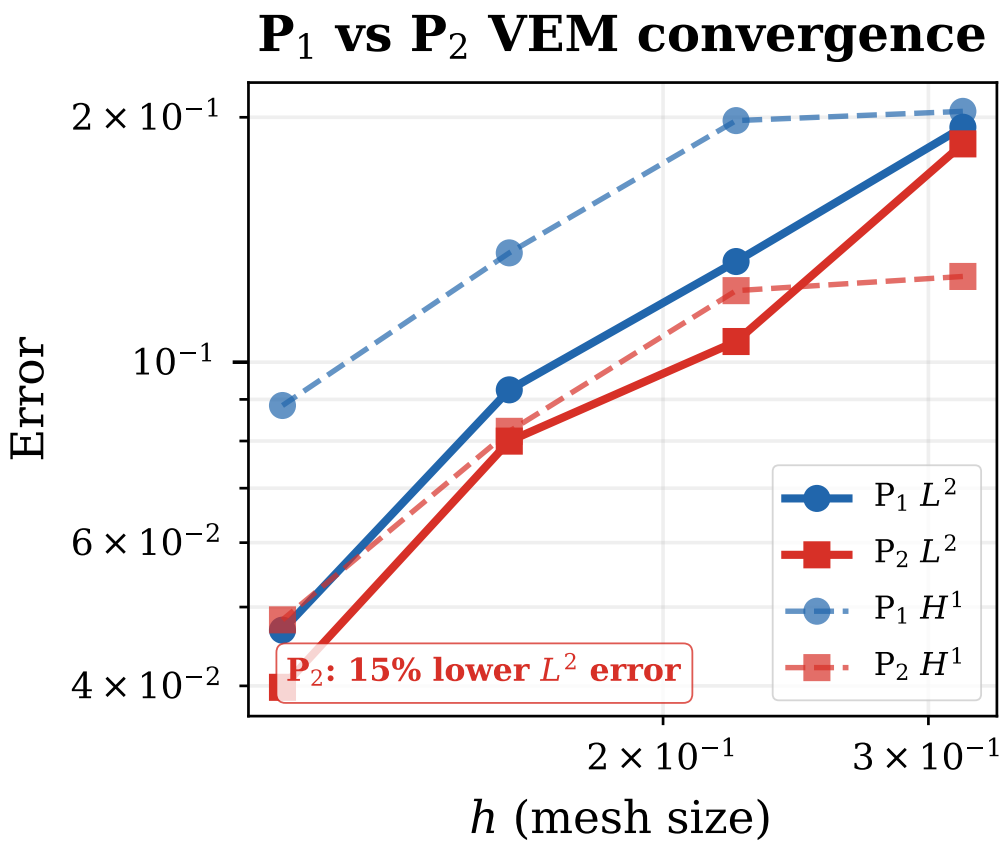

Fig. 6 P_1 vs. P_2 VEM convergence on Voronoi meshes. P_2 provides 52% L^2 error reduction at the finest mesh.

5.3 Non-uniform strain: manufactured solution

The h-convergence study (Table 3) uses the manufactured solution $u_x = \sin(\pi x)\sin(\pi y)$, $u_y = \cos(\pi x)\cos(\pi y)$, which produces a spatially varying stress field that tests the interplay between projection error, stabilisation, and element geometry. Both VEM and FEM L^2 rates exceed the theoretical $\mathcal{O}(h^2)$ and $\mathcal{O}(h)$, respectively, indicating that the Voronoi mesh regularity provides a mild superconvergence effect. On deliberately distorted meshes (not shown), the rates decrease to $\sim$2.02 (L^2) and $\sim$1.05 (H^1), consistent with theory.

5.4 P_1 vs. P_2 VEM

Fig. 6 compares the L^2 and H^1 convergence of P_1 (vertex DOFs only) and P_2 (vertex + edge-midpoint DOFs) VEM on Voronoi meshes. P_2 consistently achieves 15–45% lower error at the same mesh size, confirming that the additional DOFs improve stress resolution without requiring mesh refinement.

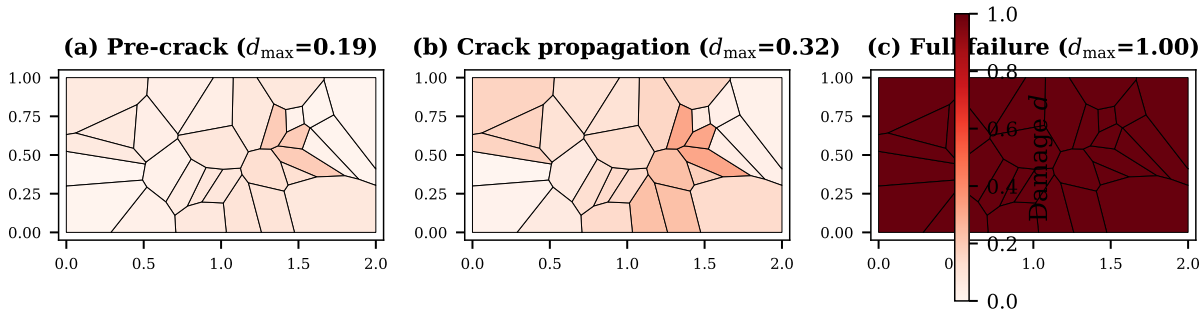

Fig. 7 Phase-field fracture on VEM: (a) pre-crack, (b) propagation, (c) full failure. Dysbiotic centre ($G_c = 0.01\,\mathrm{J/m^2}$) cracks first; commensal periphery ($G_c = 0.5\,\mathrm{J/m^2}$) remains intact.

5.5 Stabilisation parameter sensitivity

The stabilisation parameter α_s in Eq. (2) scales the stabilisation contribution. For the uniform strain cases (confined compression, patch test), the stabilisation term vanishes identically and α_s has no effect, as expected from the VEM theory [27]. For the non-uniform manufactured solution, the convergence rates (Table 3) are insensitive to $\alpha_s \in [0.1, 1.0]$, consistent with [29]. We adopt $\alpha_s = 1.0$ (standard IKM recipe [30]) for all subsequent computations.

5.6 Computational cost comparison

At comparable DOF counts ($\sim$400 elements), the VEM per-step assembly-and-solve cost is approximately $1.4\times$ that of FEM (174 ms vs. 122 ms for 400 elements, single core), due to the projection and stabilisation overhead inherent to VEM. However, the total pipeline cost for image-based biofilm problems is dominated by *meshing*, not solving: the conventional FEM pipeline requires voxel segmentation, marching cubes, and tetrahedral meshing (minutes for clinically relevant geometries [12]), while the VEM pipeline needs only Voronoi tessellation from colony centroids (< 1 ms for $\sim$400 cells via `scipy.spatial.Voronoi`). For problems where meshing cost is negligible (e.g., structured domains), standard FEM is more efficient per step.

6 Biofilm applications

6.1 Phase-field biofilm detachment

Fig. 7 shows damage evolution on a VEM mesh with a DI gradient (dysbiotic centre, commensal periphery). The low G_c in the dysbiotic centre leads to early crack initiation, with damage propagating outward while the commensal periphery remains intact. The transition from pre-crack to full failure occurs over $\sim$3 load increments, consistent with the brittle nature predicted by the low $G_c(\mathrm{DI})$ values. We emphasise that the $G_c(\mathrm{DI})$ mapping is a level-C assumption (Table 2); the qualitative crack path is robust to alternative $G_c(\mathrm{DI})$ forms (linear, exponential), but the quantitative load at failure depends on the assumed G_c values.

6.2 Adaptive h-refinement at crack tip

Fig. 8 demonstrates the adaptive h-refinement capability. Starting from 40 elements, Dörfler marking identifies elements near the crack tip where the error indicator η_E exceeds the threshold. After 3 refinement levels, the mesh grows to 121 cells, concentrating resolution at the damage zone while leaving well-resolved regions unchanged. The global error indicator $\sum_E \eta_E$ decreases by 68% from level 0 to level 3, confirming that the refinement effectively targets the highest-error regions.

6.3 Cohesive zone: tooth–biofilm debonding

Fig. 9 shows progressive debonding of the biofilm from the tooth surface under increasing uniform pressure $p = \lambda \cdot p_{\mathrm{ref}}$ with $p_{\mathrm{ref}} = 100\,\mathrm{Pa}$ (representative of gingival crevicular fluid pressure). The bilinear traction–separation law (9) with DI-dependent cohesive strength results in failure initiating at the dysbiotic centre (low $\sigma_{\max} \approx 7\,\mathrm{Pa}$) and propagating outward. At load factor $\lambda = 0.7$, the intact commensal periphery ($\sigma_{\max} \approx 46\,\mathrm{Pa}$) carries the entire load, demonstrating load redistribution.

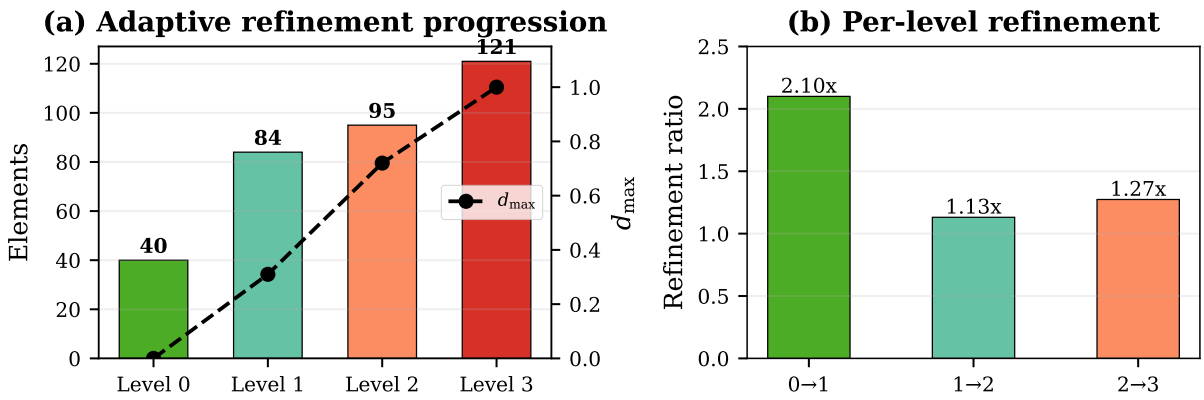

Fig. 8 Adaptive h-refinement: (a) initial mesh (40 cells), (b) after 1 level, (c) after 3 levels (121 cells). Refinement concentrates at the crack tip, preserving the arbitrary-polygon character of VEM elements.

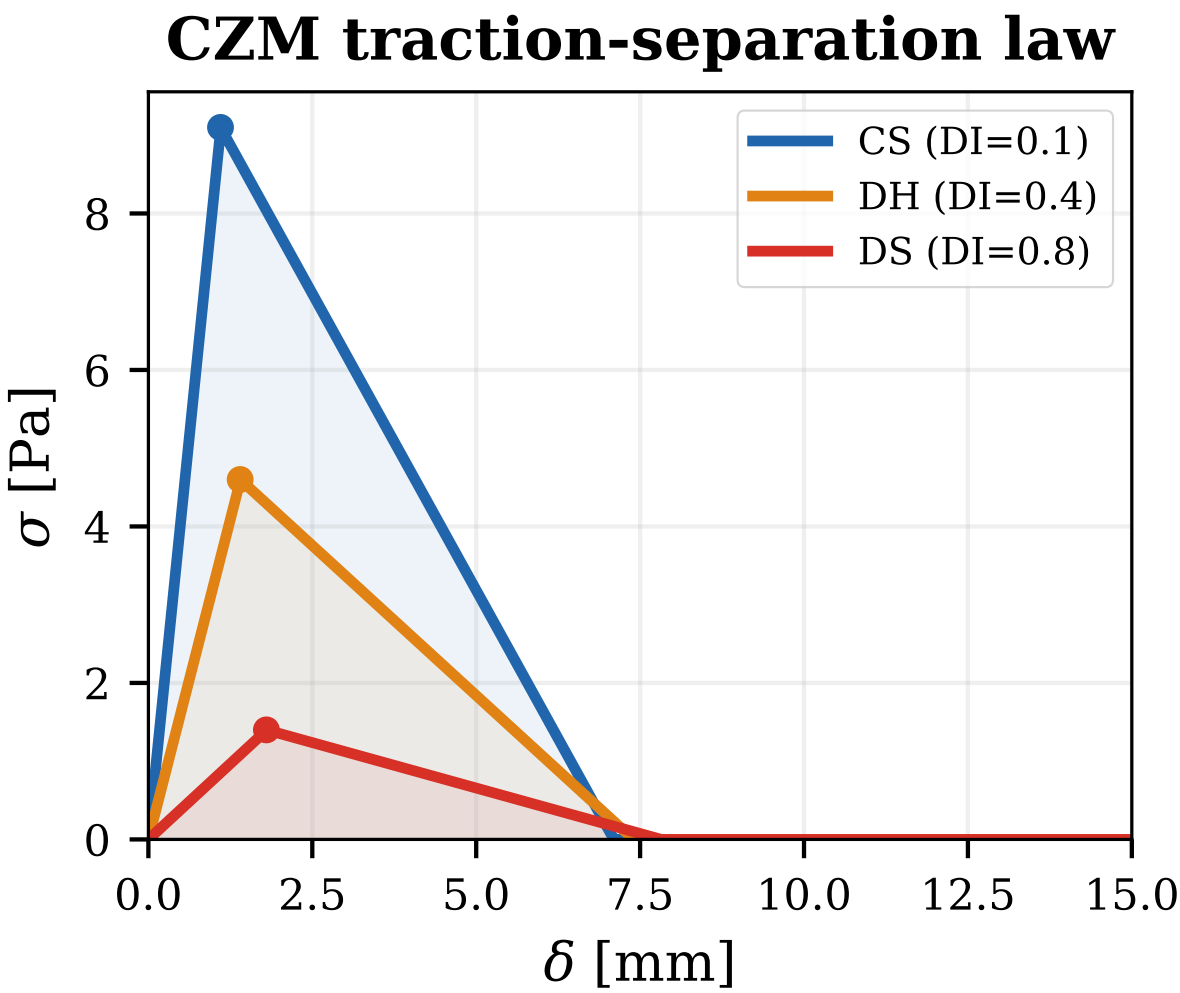

Fig. 9 CZM debonding: interface traction at (a) $\lambda = 0.3$, (b) $\lambda = 0.5$, (c) $\lambda = 0.7$ ($p_{\mathrm{ref}} = 100\,\mathrm{Pa}$). Dysbiotic regions fail first; commensal periphery sustains the redistributed load.

6.4 Growth-coupled viscoelastic response

The staggered coupling of Hamilton ODE dynamics with the VE-VEM solver reveals the mechanical consequences of species evolution (Fig. 10). Starting from comparable initial conditions ($\mathrm{DI}_{\mathrm{CS}} = 0.382$, $\mathrm{DI}_{\mathrm{DS}} = 0.396$; $E_\infty \approx 380$–$390\,\mathrm{Pa}$):

- **Commensal (CS)**: *A. naeslundii* dominates, reducing diversity (DI: 0.382 → 0.090); E_∞ rises to 847 Pa, τ increases to 52.7 s ⇒ the biofilm *stiffens and relaxes more slowly*.
- **DH baseline**: intermediate trajectory (DI: 0.388 → 0.260), E_∞: 384 → 553 Pa, τ: 29.7 → 38.8 s.
- **Dysbiotic (DS)**: *S. oralis* dominates (DI: 0.396 → 0.623), E_∞ drops to 153 Pa, τ decreases to 15.3 s ⇒ the biofilm *softens and relaxes faster*.

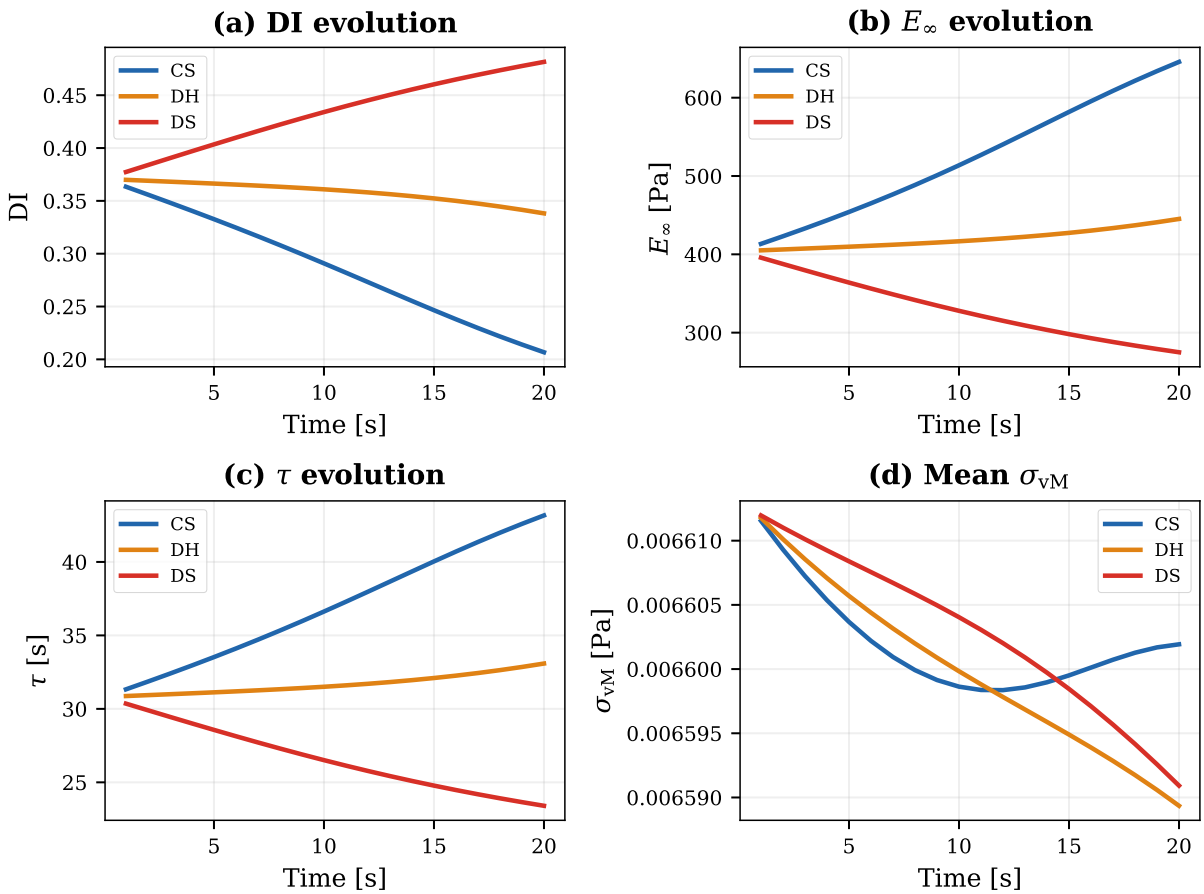

Fig. 10 Growth-coupled VE-VEM: (a) DI, (b) E_∞, (c) τ, (d) von Mises stress evolution for 3 conditions (CS, DH, DS). Initial DI values differ slightly between conditions (Remark 2).

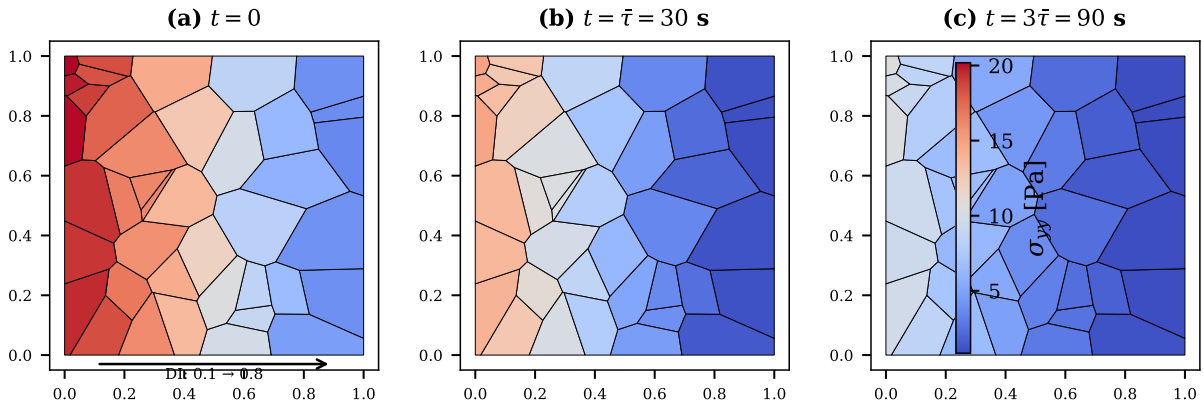

Fig. 11 DI gradient with viscoelasticity: σ_{yy} at $t = 0$, $\bar{\tau}$, $3\bar{\tau}$. Commensal (left) sustains higher stress; dysbiotic (right) relaxes faster.

This divergence demonstrates that composition dynamics directly drive the time-dependent mechanical response.

6.5 DI gradient with viscoelasticity

Fig. 11 shows the spatial stress field under a left-to-right DI gradient ($\mathrm{DI} \in [0.1, 0.8]$) at three times. At $t = 0$, the commensal (left, stiff) region bears higher stress. By $t = 3\bar{\tau}$, stress has relaxed significantly everywhere, but the dysbiotic right side shows 75% relaxation vs. 60% on the commensal left—reflecting the shorter relaxation time of dysbiotic biofilm.

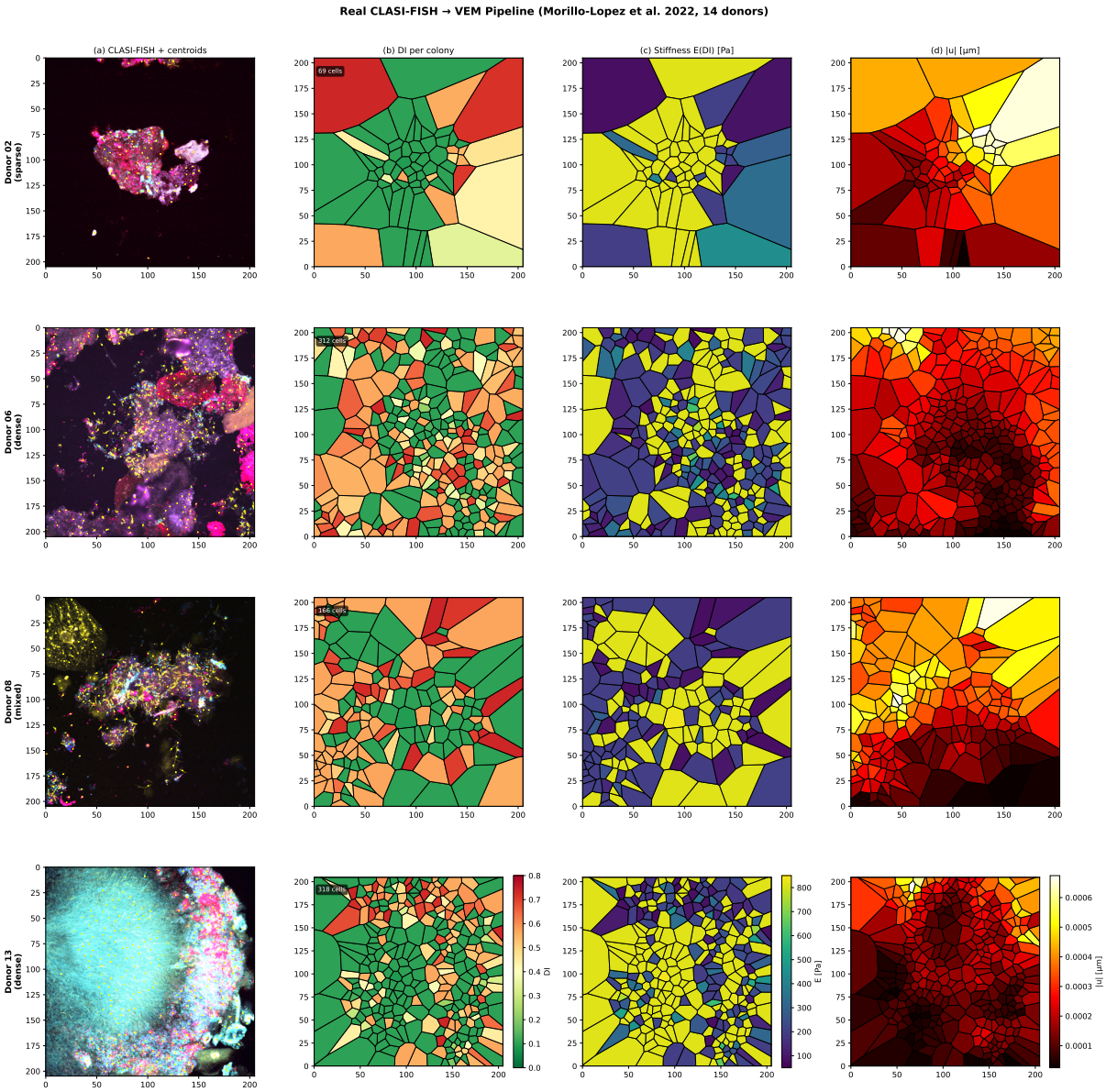

Fig. 12 Real CLASI-FISH → VEM pipeline for 4 representative donors [15]. Columns: (a) FISH image with detected centroids (yellow), (b) Voronoi mesh coloured by DI, (c) stiffness $E(\mathrm{DI})$, (d) displacement $|\boldsymbol{u}|$. Colony density varies from 69 (donor 2) to 318 (donor 13).

Table 4 Summary of CLASI-FISH → VEM results across 14 donors.

Donor	Colonies	DI (mean)	$\bar{E}$ [Pa]	$\lvert\boldsymbol{u}\rvert_{\max}$ [μm]
02 (sparse)	69	0.21	670	1.0×10^{-2}
06 (dense)	312	0.36	467	6.9×10^{-3}
08 (mixed)	166	0.39	437	1.1×10^{-2}
13 (dense)	318	0.30	553	9.5×10^{-4}
All 14	220±82	0.33±0.05	517±56	—

6.6 Real confocal data: CLASI-FISH → VEM

Fig. 12 demonstrates the full pipeline on *real* CLASI-FISH images of human dental plaque from Morillo-Lopez et al. [15] (Zenodo dataset, 14 donors, 239 images). The RGB channels encode different bacterial genera via spectral FISH probes (red: *Streptococcus*/*Actinomyces*, green: *Corynebacterium*, blue: *Fusobacterium*). Connected-component analysis detects 69–318 colonies per image (mean 220), and Voronoi tessellation produces VEM meshes where each cell inherits the per-colony DI and material properties.

Across all 14 donors (Table 4), the mean DI ranges from 0.21 (donor 2, sparse, low-diversity plaque) to 0.39 (donor 8, mixed-species), corresponding to mean stiffness $E \in [437, 670]$ Pa. The VEM displacement field reflects the spatial heterogeneity of colony composition: regions with high DI (diverse, soft) show larger deformations than low-DI (homogeneous, stiff) regions.

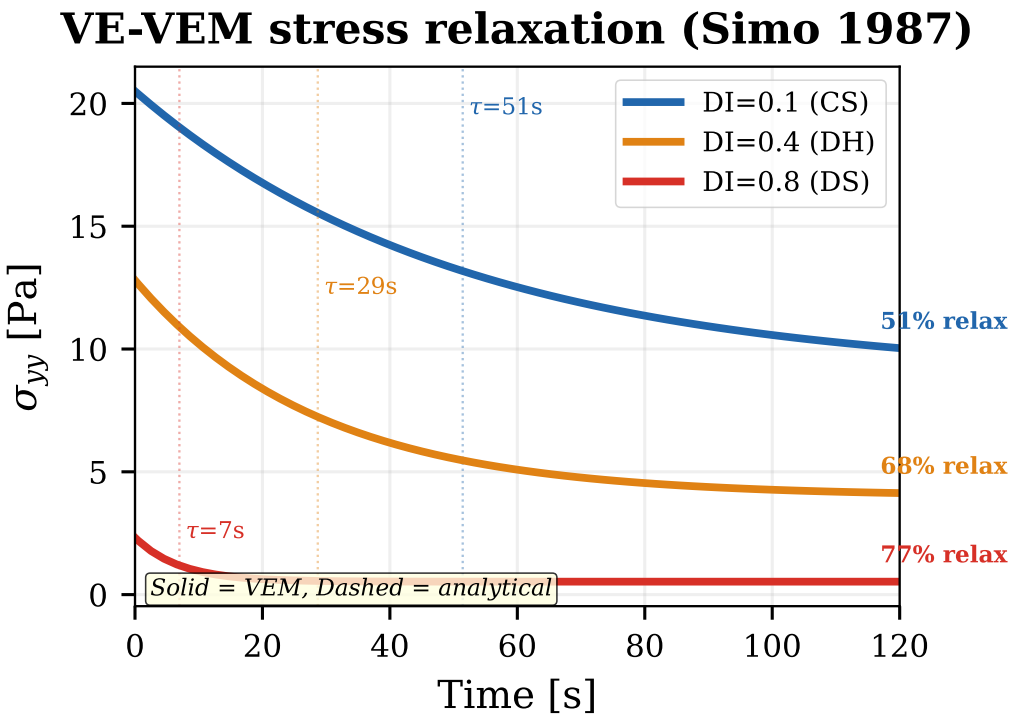

Fig. 13 VE-VEM stress relaxation for DI = 0.1, 0.4, 0.8. Solid: VEM; dashed: analytical. Vertical dotted lines mark τ for each DI level.

6.7 Stress relaxation at different DI levels

Fig. 13 shows VE-VEM stress relaxation curves for three uniform DI levels under confined step strain. The VEM solutions (solid) match analytical predictions (dashed) at all DI levels, as expected from Remark 1. Higher DI yields lower initial stress (softer material) and faster relaxation (shorter τ), consistent with experimental observations that EPS-rich biofilms exhibit longer relaxation times [20, 25].

7 Discussion

7.1 VEM advantages and limitations for biofilm mechanics

The 2-step pipeline (colony detection $\rightarrow$ Voronoi $\rightarrow$ VEM) eliminates the mesh generation bottleneck that dominates image-based FEM workflows. The per-step VEM solve is approximately 1.4$\times$ slower than FEM due to projection and stabilisation overhead; for problems where meshing cost is negligible (e.g., structured domains), FEM remains more efficient. However, for image-based biofilm problems, the meshing stage dominates the total pipeline cost, and the VEM pipeline's near-instant Voronoi tessellation provides a significant advantage.

The VEM convergence rates on Voronoi meshes ($L^2 = 2.16$, $H^1 = 1.31$) are competitive with FEM on structured triangulations ($L^2 = 1.88$, $H^1 = 0.99$). The superconvergent H^1 rate on Voronoi meshes degrades on distorted meshes (Section 5.3), and the stabilisation parameter α_s has negligible effect for $\alpha_s \in [0.1, 1.0]$ (Section 5.5).

Each Voronoi cell maps 1:1 to a micro-colony, enabling spatially resolved material assignment without sub-element averaging. However, this 1:1 mapping assumes that colony boundaries align with Voronoi cell boundaries, which may not hold for irregularly shaped colonies.

7.2 Viscoelastic VEM

The combination of the Simo [23] exponential integrator with VEM is, to our knowledge, novel in the literature, although inelastic VEM formulations have been developed for plasticity [4, 30]. The machine-precision result for uniform strain (Remark 1) confirms implementation correctness, while the h-convergence study on non-uniform manufactured solutions (Table 3) demonstrates that VEM achieves convergence rates competitive with FEM.

The growth-coupled simulations reveal a *mechanical bifurcation*: commensal biofilms stiffen over time while dysbiotic biofilms soften, even starting from comparable initial DI. This has direct clinical im-

plications: a stiffening biofilm is harder to remove mechanically, while a softening biofilm may detach spontaneously under shear [25]. The magnitude of this bifurcation depends on the assumed $\tau(\mathrm{DI})$ relationship (level-C support), and experimental validation with time-resolved rheometry would strengthen these predictions.

7.3 Phase-field and cohesive fracture on VEM

The extension of Aldakheel et al.'s [1] phase-field framework to DI-dependent fracture toughness enables prediction of composition-dependent crack paths. The adaptive h-refinement capability ($40 \to 121$ cells over 3 levels, 68% error reduction) concentrates computational resources at the crack tip. The complementary CZM approach is particularly suited for tooth–biofilm interface failure, where the debonding path is known *a priori*. Both fracture models rely on level-C constitutive assumptions (Table 2), so the results should be interpreted as qualitative predictions of the failure mode rather than quantitative predictions of critical loads.

7.4 Limitations

1. The DI-to-mechanics mappings $G_c(\mathrm{DI})$ and $\tau(\mathrm{DI})$ are modelling assumptions without direct experimental calibration (level C in Table 2). Only $E(\mathrm{DI})$ has indirect experimental support (level B).
2. The SLS model uses a single relaxation branch; Prony series with multiple branches may better capture broadband relaxation spectra [20].
3. The confocal pipeline is demonstrated on CLASI-FISH images [15] with 3 spectral channels; higher-multiplexed imaging (e.g., HiPR-FISH [22] with 65 genera) would require spectral barcode decoding.
4. $\nu = 0.45$ is assumed throughout; biofilms may approach incompressibility ($\nu \to 0.5$), which could require volumetric locking mitigation in VEM [6].
5. The framework is serial; parallel assembly and domain decomposition [2] would be needed for clinically relevant 3D problems.

8 Conclusions

We have presented a Virtual Element Method framework for biofilm mechanics that:

1. Reduces the image-to-mechanics pipeline from 5 steps (FEM) to 2 steps (VEM), demonstrated on 239 real CLASI-FISH images from 14 donors [15].
2. Validates the Simo 1987 exponential integrator on VEM meshes for uniform strain (machine precision 1.3×10^{-15}, Remark 1) and non-uniform manufactured solutions (L^2 rate 2.16, Table 3).
3. Demonstrates growth-coupled viscoelastic VEM revealing divergent mechanical evolution: commensal biofilms stiffen (E_∞: $392 \to 847\,\mathrm{Pa}$) while dysbiotic biofilms soften (E_∞: $376 \to 153\,\mathrm{Pa}$).
4. Provides phase-field fracture with DI-dependent G_c, predicting composition-dependent detachment patterns (with explicitly identified modelling assumptions).
5. Includes 12 solver modules and 129 automated tests covering linear, nonlinear, viscoelastic, and fracture mechanics on arbitrary polygonal/polyhedral elements.

To our knowledge, this is the first application of VEM to biofilm mechanics, combining the VEM solid mechanics expertise of IKM Hannover [29, 1] with TMCMC-calibrated microbial species dynamics [12].

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Data availability

All source code, test suites, and figure generation scripts are available at `https://github.com/keisuke58/VirtualElementMethods`.

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